

Class Meeting Handout #8

Barry Linkletter

More very important review. Kinetic analysis will allow us to investigate reaction mechanism and make hypotheses for how to change conditions or reactants so that we can make our synthetic dreams come true.

Reaction Kinetics

If the kinetics of a reaction fit the mathematical model of the rate law then the hypothesis might be correct. Being able to propose a rate law and fit the observed data to the model is an important tool in physical organic chemistry.¹

Assignment Discussion: Our Core Values – 10 minutes

Let us review what you have learned during the assignment activity.

1. Describe the Hammond postulate, the principle of microscopic reversibility, the Curtin-Hammett principle, the idea of thermodynamic and kinetic control and the reactivity-selectivity principle.²

Review: Kinetics Roundup – 40 minutes

Now lets refresh our reaction kinetics skills.

2. Write out single step reactions that can be described as first and second order. Propose a rate law for each and integrate the rate law to obtain a model that describes the concentration of reactants and products over time.³
3. The data in table 1 describes the decrease in reactant concentration over time. Determine the order of reaction and the rate constant.⁴
4. For a reaction of the form $nA \longrightarrow B$, show that a plot of $\log(v_{t=0})$ vs. $[A]$ will have a slope equal to the order of the reaction, n .⁵
5. Identify the rate-determining step in the cases shown in figure 1.
6. *t*-Butyl bromide can be converted to *t*-butanol in boiling water. Answer the following questions.
 - (a) Propose a mechanism for this reaction. Be sure that it is consistent with all observations below.
 - (b) The hydrolysis of *t*-butyl bromide is slower when excess concentrations of NaBr are present. Explain this observation.
 - (c) Use the steady-state approximation method to derive the rate law.⁶

This document was produced using the L^AT_EX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in ChemDoodle,

¹ Read chapter 7 of the textbook.

² Ch. 7.3

Table 1: Concentration vs. time data for a reactions.

time (/min)	Reactant (/M)
0	0.250
0.042	0.213
0.092	0.175
0.119	0.156
0.151	0.138
0.280	0.081
0.435	0.044

³ See the tutorial on integrating rate laws available from the class moodle site page for this meeting.

⁴ Ch. 7.4

⁵ $v_{t=0}$ is the rate at $t = 0$. This is an example of the initial rate method.

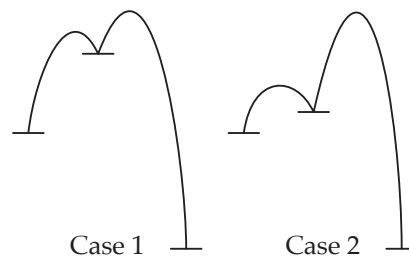


Figure 1: Example reaction coordinates for a two-step reaction.

⁶ Ch. 7.5

7. The hydration of isobutene is faster in systems that are more acidic. Answer the following questions.

- Draw a mechanism for hydration of an alkene in acid and identify the rate-determining step.
- The rate of the reaction is nearly identical in 2 M HCl and in 2 M HI. The HI acid is 10^3 -fold stronger acid compared to HCl.⁷ Explain.
- If we measure the rates in acid mixtures with increasing acidity, we observe a first-order dependence in acid activity.^{8,9} We can say that $k_{obs} \propto a_{H^+}$. Derive a rate law that is consistent with your mechanism.

8. *t*-Butanol can be converted to *t*-butyl chloride in the presence of HCl. Answer the following questions.

- Draw a mechanism for S_N1 substitution reaction of *t*-butanol in the presence of HCl and identify the rate-determining step.
- This reaction is first order in acid activity for conditions where $H_0 > -2$. As we increase acidity the reaction becomes zero-order in acid. Draw a plot of $\log k_{obs}$ vs H_0 and explain this observation.
- Does the concentration of acidic species change during this reaction. Does it fit the definition of a catalyst?
- Derive a rate law for the reaction using the pre-equilibrium assumption.⁶
- The H_0 scale is based on observation with strong acids protonating substituted anilines. The H_R scale^{10,11} is similarly based on a series of substituted triaryl carbinols. Should we use the H_R scale instead?^{12,13}

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the moodle site you will have achieved the following...

- Identify the likely rate-determining step in a reaction and construct a rate law from a reaction scheme.
- Design an experiment to investigate reaction kinetics in a given reaction
- Construct a rate law from a complex reaction scheme using the pre-equilibrium assumption of steady-state assumption.
- Be able to interpret data from reaction kinetics experiments.

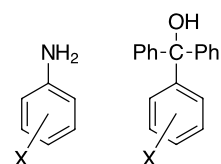
Looking Ahead

In the final question today we considered an example of an acid catalyzed reaction. Acid and base catalysis is what the next three class meetings will explore. We're ready now, let's go!

⁷ See the data collection curated by the ACS Organic Chemistry Division at <https://organicchemistrydata.org/hansreich/resources/pka/> There is much more there than pK_a values so be sure to look around.

⁸ "The Dependence of the Rate of Hydration of Isobutene on the Acidity Function, H_0 , and the Mechanism for Olefin Hydration in Aqueous Acids." R.W. Taft Jr. *J. Am. Chem. Soc.*, 1952, 74, 5372-5376. <https://doi.org/10.1021/ja01141a046>

⁹ "The mechanisms of isobutene hydration yielding tert-butanol catalyzed by a strong mineral acid (H_2SO_4) and Lewis-Brønsted superacid (HF/SbF_5).” J. Brzeski, P. Skurski, *Heliyon*, 2019, 5, e02133. <https://doi.org/10.1016/j.heliyon.2019.e02133>



Aniline Triaryl carbinol

¹⁰ "The ionisation of triarylcarbinols in strong acids and the definition of a new acidity function" V. Gold, B.W.V. Hawes, *J. Chem. Soc.*, 1951, 2102-2111. <https://doi.org/10.1039/JR9510002102>

¹¹ "Carbonium Ions. VIII. Diarylolefin-Diaryllalkyl Cation Equilibria" N.C. Deno, P.T. Groves, G. Saines, *J. Am. Chem. Soc.*, 1959, 81, 5790-5795. <https://doi.org/10.1021/ja01530a059>

¹² For a useful review see "Acidity functions: an update." R.A. Cox, K. Yates, *Can. J. Chem.*, 1983, 61, 2225-2243. <https://doi.org/10.1139/v83-388>

¹³ For more examples of acidity function scales, see "Protonation and Solvation in Strong Aqueous Acids." E.M. Arnett, G. Scorranno, *Adv. Phys. Org. Chem.*, 1976, 13, 83-153. [https://doi.org/10.1016/S0065-3160\(08\)60213-0](https://doi.org/10.1016/S0065-3160(08)60213-0). Accessed via inter-library loan. Ask me for a copy.